NETFLOW

SOLID MATERIAL LEVEL CONTROL · SERIES

ROTARY PADDLE LEVEL SWITCH

Rotary paddle level sensor for full / empty detection in silos, bunkers and containers

The series rotary paddle level switch mechanically detects the level of powder and granular solids in silos. The paddle, driven at 5 rpm by a geared synchronous motor, is stopped when covered by material; the resulting reaction movement of the motor body is detected by a limit switch (micro switch), a relay contact output is generated and the motor stops. Detection sensitivity can be set in 3 steps according to the bulk density of the material.

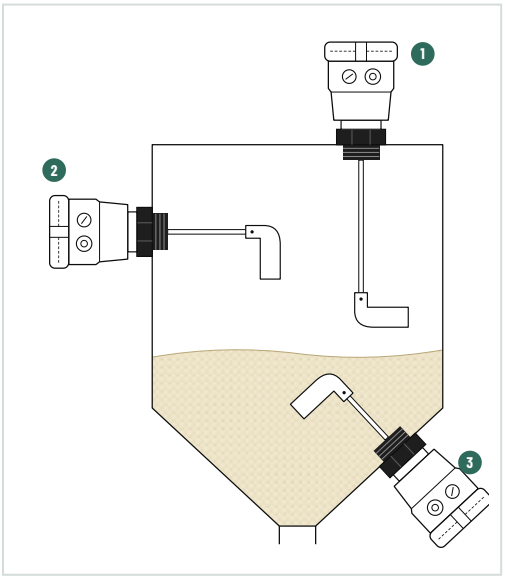


· L TYPE PADDLE

IP66 / IP67 PROTECTION CLASS	**3 Steps** SENSITIVITY SETTING
24 VDC / 220 VAC SUPPLY OPTIONS	**5 rpm** PADDLE SPEED

- 100 / 200 mm shaft standard, 200 / 500 / 1000 mm extension shaft optional
- Limit switch / micro switch detection
- SS316 stainless steel shaft and paddle
- 1½" BSPP connection, 1" and 2" optional
- L and T type paddle, selected per application
- Glass-fibre reinforced PC-ABS connection; aluminium and stainless optional

TYPICAL USE



1 VERTICAL MOUNTING · HIGH LEVEL

Mounted from the silo roof. Used for full-level alarm and filling stop; the switching point is set by the shaft length.

2 SIDE MOUNTING · INTERMEDIATE LEVEL

Mounted horizontally on the silo wall. For high or intermediate level control, positioned outside the filling stream.

3 ANGLED MOUNTING · LOW LEVEL

Mounted at an angle on the cone. Used for empty alarm and feed demand; the angled position reduces material build-up on the paddle.

APPLICATION AREAS



CEMENT · READY-MIX CONCRETE



GYPSUM · LIME · AGGREGATE

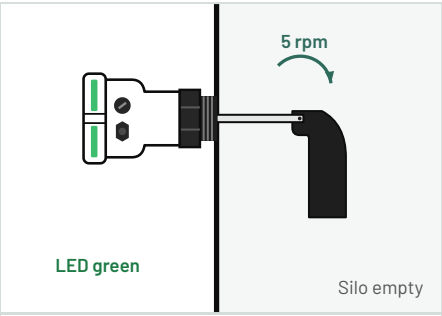


FEED · SEED · FLOUR



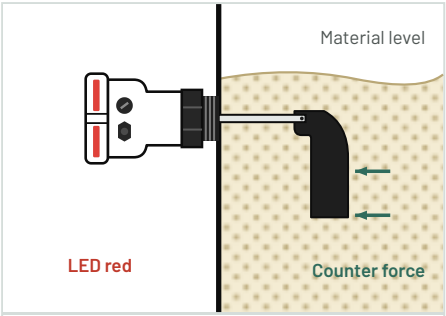
PLASTICS · COAL DUST

OPERATING PRINCIPLE



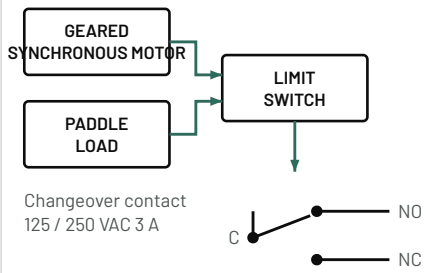
1 EMPTY STATE

The geared synchronous motor rotates the paddle shaft continuously at 5 rpm. As long as the shaft turns freely, the relay stays in the empty state and the indicator LED is green.



2 MATERIAL CONTACT

When material covers the paddle, a counter force acts on it. This force rises until it exceeds the threshold set by the 3-step spring adjustment.



3 SIGNAL OUTPUT

The load is detected by the limit switch; the motor stops and the changeover contact switches within max. 0.6 s, the indicator LED turns red. When the material recedes, the system resets itself.

TECHNICAL SPECIFICATIONS

ELECTRICAL DATA

Supply (24 VDC)	24 VDC ±3%, max. 2.7 W
Supply (220 VAC)	220 VAC, max. 3.1 W
Signal output	1 changeover contact (C / NO / NC), 24 VDC / 220 VAC 3 A
Signal delay	Max. 0.6 s
Terminal	Max. 2 mm² (AWG 14) cable
Electrical connection	PG 9 cable gland · optional DIN 43650 (ABS) connector or M12 4-pin
Status indicator	LED · empty: green · full: red
Protection class	IP66 / IP67

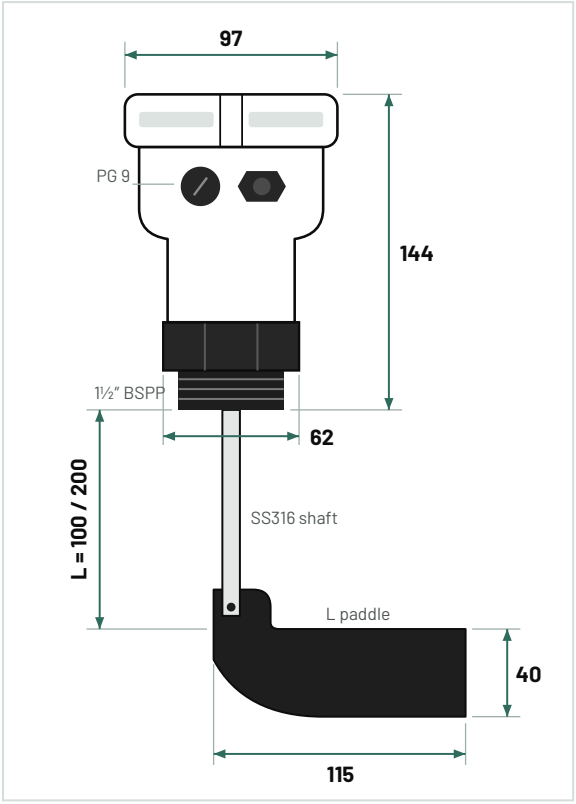
OPERATING CONDITIONS

Ambient temperature	–20 °C ... +80 °C
Process temperature	–5 °C ... +80 °C · +160 °C with cooling extension
Silo pressure	–0,5 ... +0,5 bar
Max. operating vibration	5 – 500 Hz, 3G RMS (IEC 60068-2-64)

MECHANICAL DATA

Housing	Option P: PC-ABS · Option A: machined aluminium
Drive	Geared synchronous motor
Detection	Limit switch (micro switch)
Shaft bearing	Double ball bearing
Sealing	NBR seal (Viton optional)
Connection material	Glass-fibre reinforced PC-ABS · aluminium and stainless optional
Connection size	1½" BSPP (standard) · 1" and 2" BSPP optional
Shaft and paddle	Stainless steel SS316
Shaft length	100 mm (standard) · 200 mm · optional 200 / 500 / 1000 mm extension shaft
Shaft speed	5 rpm
Sensitivity setting	3-step spring
Min. bulk density	250 g/l (with standard L paddle)
Weight	870 g (100 mm shaft) · 1000 g (200 mm shaft), excl. extension shafts

DIMENSIONS
DIMENSIONS mm



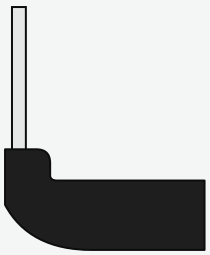
CONNECTION SIZES



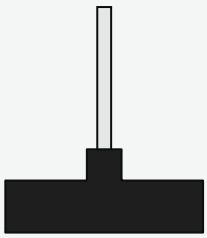
1" BSPP	1½" BSPP	2" BSPP
OPTIONAL	STANDARD	OPTIONAL

Shaft length 100 mm (standard) or 200 mm. For deeper switching points, 500 / 1000 mm extension shafts are available as options.

PADDLE TYPES



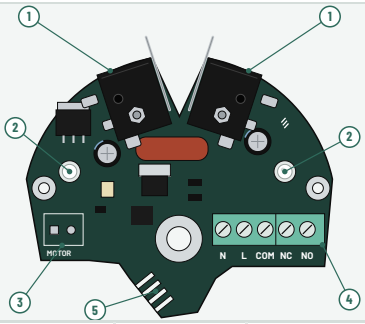
L TYPE
Standard. Powders and fine granules, density above 250 g/l.



T TYPE
Large surface. Earlier detection in light and fluffy materials. Single blade, rod-connected spring system.

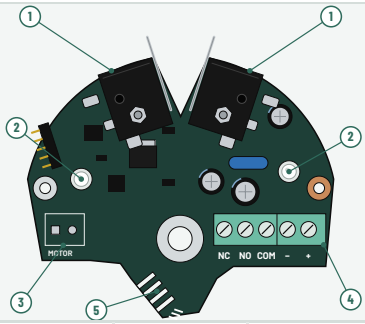
ELECTRICAL CONNECTION
ELECTRONIC BOARD

220 VAC MODEL
Supply: terminals 1 – 2



N	L	COM	NC	NO
1 · Neutral	2 · Phase	3 · Common	4 · N. closed	5 · N. open

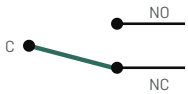
24 VDC MODEL
Supply: terminals 4 – 5



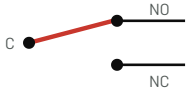
NC	NO	COM	–	+
1 · N. closed	2 · N. open	3 · Common	4 · 0 V	5 · 24 VDC

(1) **Limit switch** 2 pcs, load detection
(2) **LED** Empty: green · Full: red
(3) **Motor** Motor connection
(4) **Terminal block** 5-pole, dry contact
(5) **Spring step** Torque setting

SIGNAL STATUS



SILO EMPTY
C – NC closed · LED green · paddle rotates



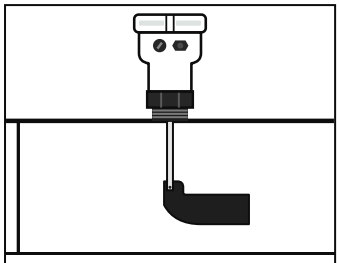
SILO FULL
C – NO closed · LED red · motor stops

CONNECTION TYPES

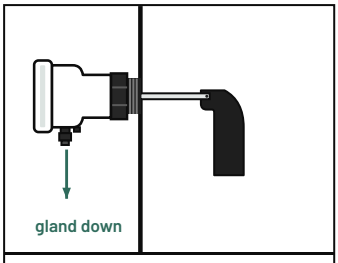
PG 9 cable gland	Standard
DIN 43650 (ABS) connector	Optional
M12 4-pin connector	Optional

Electrical connections must be made by a qualified electrician with the power off. Max. 2 mm² cable can be connected to the terminals.

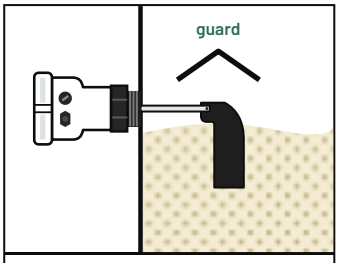
MECHANICAL INSTALLATION



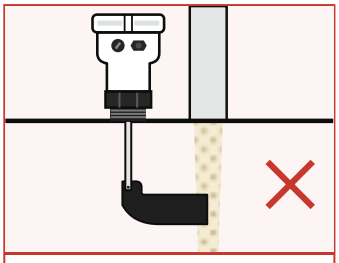
✓ Vertical from roof, perpendicular to surface



✓ In horizontal mounting the gland faces down



✓ At low level the paddle is protected by a guard



✗ Do not mount below the filling stream

01 The unit must be mounted perpendicular to the surface, away from the material inlet and filling stream.

02 The paddle must not touch the silo wall, and material must not jam between them.

03 In horizontal mounting, the cable gland or connector must face down to prevent water ingress.

04 In low-level applications and with heavy materials, a guard must be fitted above the paddle.

05 After installation, tighten the housing cover and gland fully to ensure sealing.

06 Do not mount on vibrating cones or vibrating surfaces.

TROUBLESHOOTING

SYMPTOM	POSSIBLE CAUSE	REMEDY
Shaft not rotating, LED off	No supply or wrong wiring	Measure the supply voltage at terminals 1 – 4 for 24 VDC and 1 – 4 for 220 VAC with PG 9 connection; at the relevant pins for M12 and DIN connectors.
Full signal with empty silo	Material jammed between paddle and silo wall	Clean the paddle, increase the mounting distance or add a guard.
No signal with material present	Material density below 250 g/l or spring step too high	Lower the spring step according to material density; if needed, change to a T type paddle.
Moisture / water in housing	Gland facing up or cover not fully closed	Turn the gland downwards, check the cover and seal.
Signal flickering	Mounted on a vibrating surface	Move the unit away from the vibrating cone.

SELECTION GUIDE

HOUSING

A plastic housing is sufficient for indoor and normal dusty applications. For outdoor, wash-down or impact-risk lines, an aluminium housing is preferred.

SHAFT LENGTH AND PADDLE

For materials that stick to the silo wall, choose a 200 mm shaft or an extension shaft. For light materials below 250 g/l, use a T type paddle.

HOT PROCESSES

Where the process temperature exceeds +80 °C, use the model with cooling extension (up to +160 °C). Stainless steel is recommended as connection material.

OPTIONS

- Aluminium housing (machined)
- Cooling extension model, +160 °C process
- 200 / 500 / 1000 mm extension shaft
- T type paddle
- 1″ and 2″ BSP connection
- Aluminium or stainless connection
- DIN 43650 (ABS) or M12 4-pin connector
- Viton seal

COMPLIANCE

ISO 9001 · EN 61000-6-4 · EN 61000-6-2 · EN 61010-1

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Contact our technical team to select the right shaft length, paddle type and connection size for your application.

YERLİ ÜRETİM